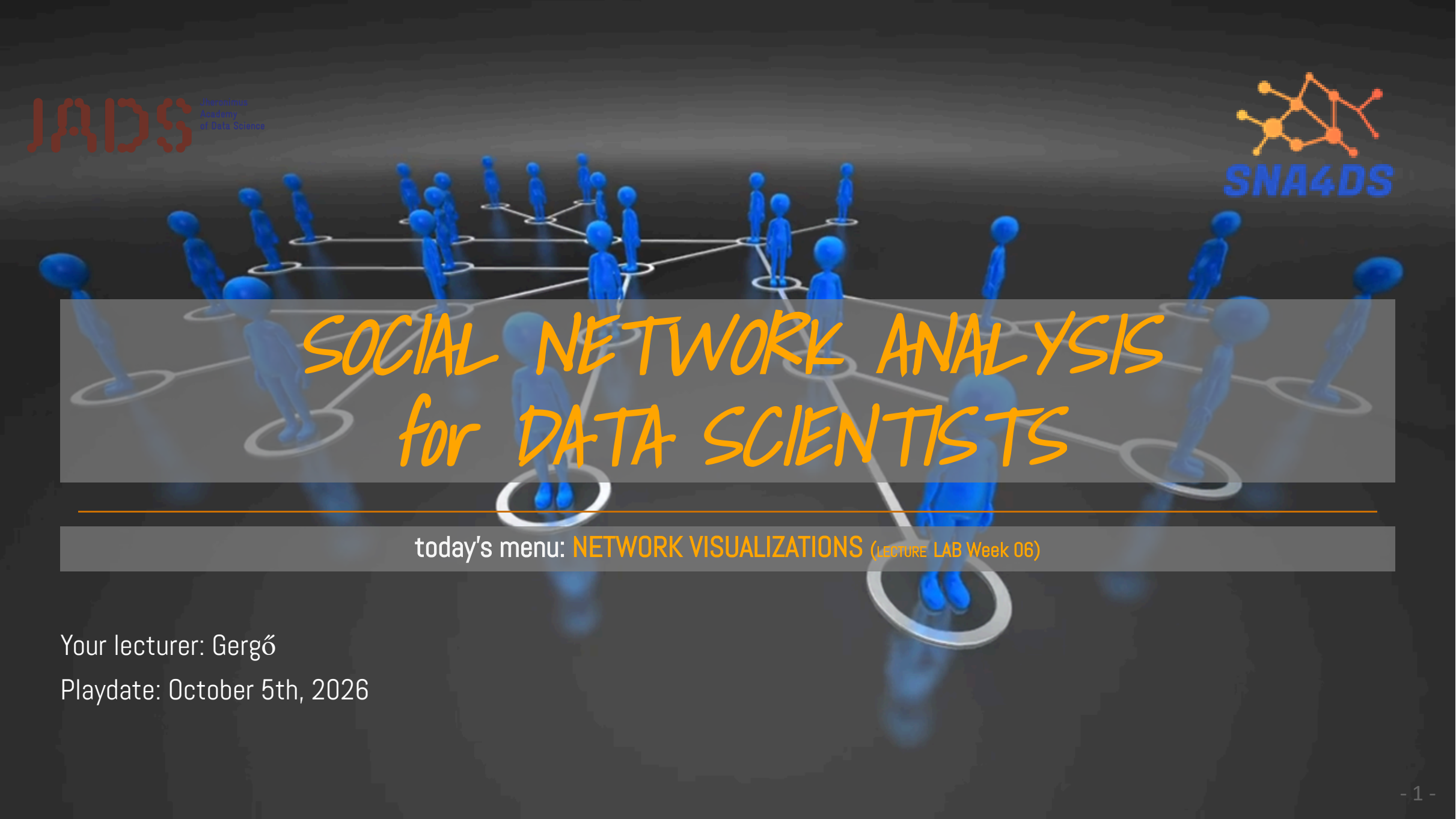


SOCIAL NETWORK ANALYSIS for DATA SCIENTISTS

today's menu: NETWORK VISUALIZATIONS (LECTURE LAB Week 06)

Your lecturer: Gergő

Playdate: October 5th, 2026

Menu' for today

1. In the making of a viz
    1. What
    2. Why
    3. How
2. Homeplay
3. Tricks or treats
4. Final Remarks

> "As network researchers our biggest advantage in communicating our impact is the really pretty pictures we can make."
> - *Anonymous Network Scholar at a conference*

1. In the making of a viz

What's a viz?



What's a viz?

VISUALIZATION = MESSAGE

Metaphor for some parts of reality or concepts that we are explaining

Anatomy of a visualization

What are you working with?

Why are you making the viz?

How can you best achieve this goal?

Excellent book on visualizations: Munzner, T. (2015). Visualization analysis & design. CRC Press. https://doi.org/10.1201/b17511

What are you working with?

What are you working with?

Taking stock of your resources

What are you working with?

Taking stock of your resources

You have a network

What are you working with?

SNA4DS

Taking stock of your resources

You have a network

What about attributes?

- Nodes/edges?
- Categorical? Continuous? Ordinal?
- Temporal stuff?
- Missing data?

What are you working with?

Taking stock of your raw materials is the foundation of the visualization process.

Why are you making the visualization?

SNA4DS

What's the goal of the viz?

What is the message you want to get across?

Who is your audience?

Why are you making the visualization?

Discover? Present? Enjoy?

Why are you making the visualization?

SNA4DS

Discover? Present? Enjoy?

Compare? Identify? Summarize?

Why are you making the visualization?

SNA4DS

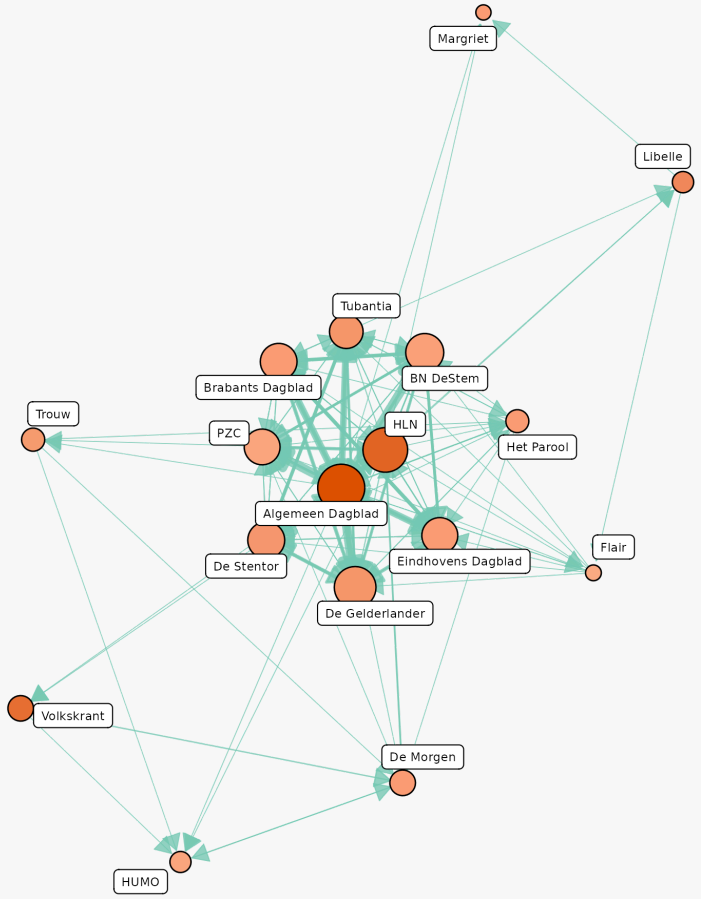
Discover? Present? Enjoy?

Compare? Identify? Summarize?

No right or wrong, just make sure you're clear on this.

Some skin in the game: my visualizations

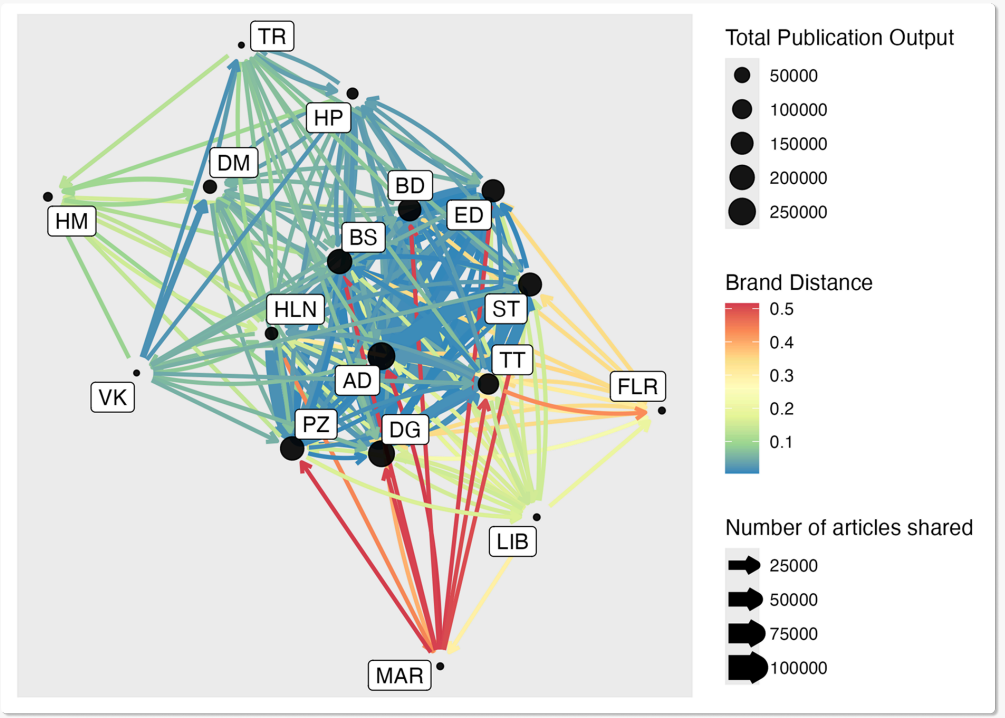
Content Sharing Network in 2023 August



Articles shared
1000
2000
3000

Number of Articles Published
5000
10000

New Subsrcibers



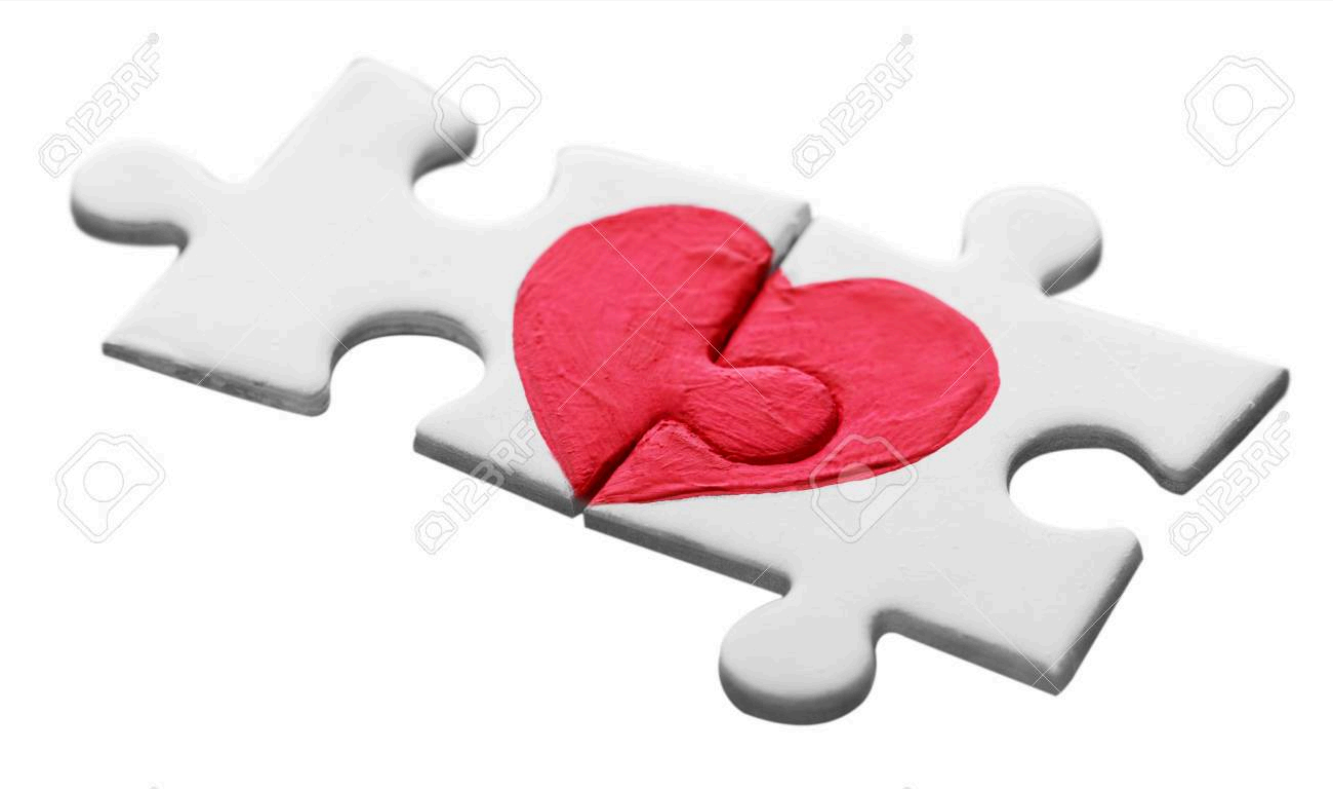
How you can best achieve these goals?



In this class it's `igraph`. Still, which `igraph` functions are most useful for your goals?

Get to know your tool, make the most of it.

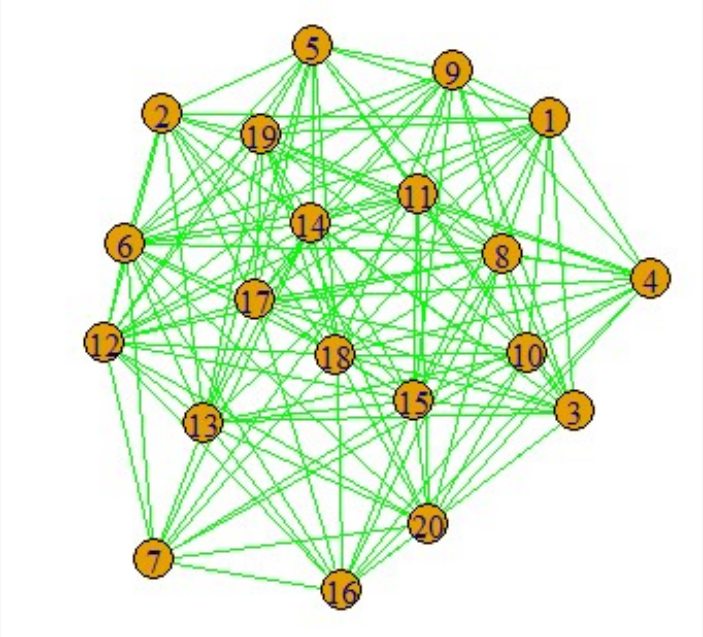
c. Does your strategy work with your data and with your tool?



Example 1

You want to show passages in a network of football players.

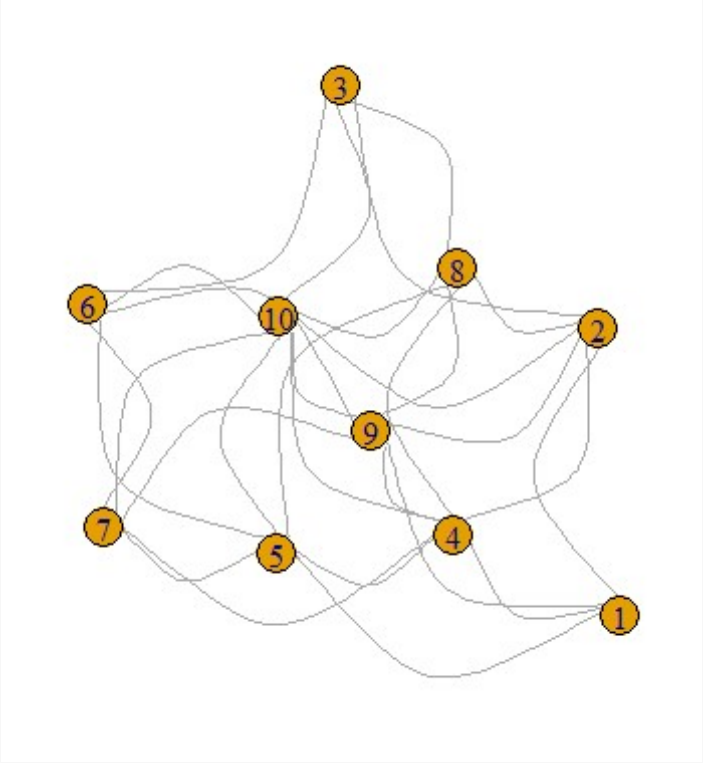
Since the football court is green you decide it would be impactful to use the color green for the edges (passages).



Match the idea with the reality of the tools you have: It is theoretically a nice, but impossible to read it.

Example 2

You are plotting a network of dancers and you decide that using curved edges is more romantic.



Match the idea with the reality of the tools you have: You might be artistically right, but it gets really hard to see where the edges are going

d. Fancy versus functional

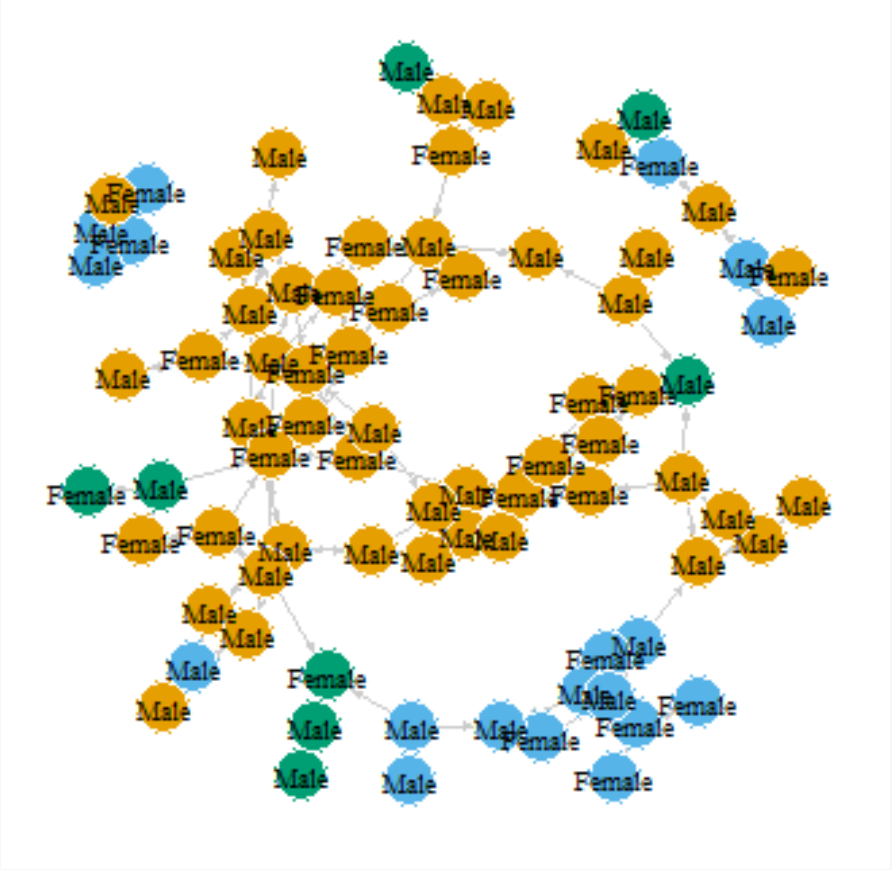
SNA4DS

You want to insert too much information in a very elaborated way

- many colors
- weird node shapes
- labels on the nodes and edges
- …

Networks are a bit too powerful for their (your) own good.

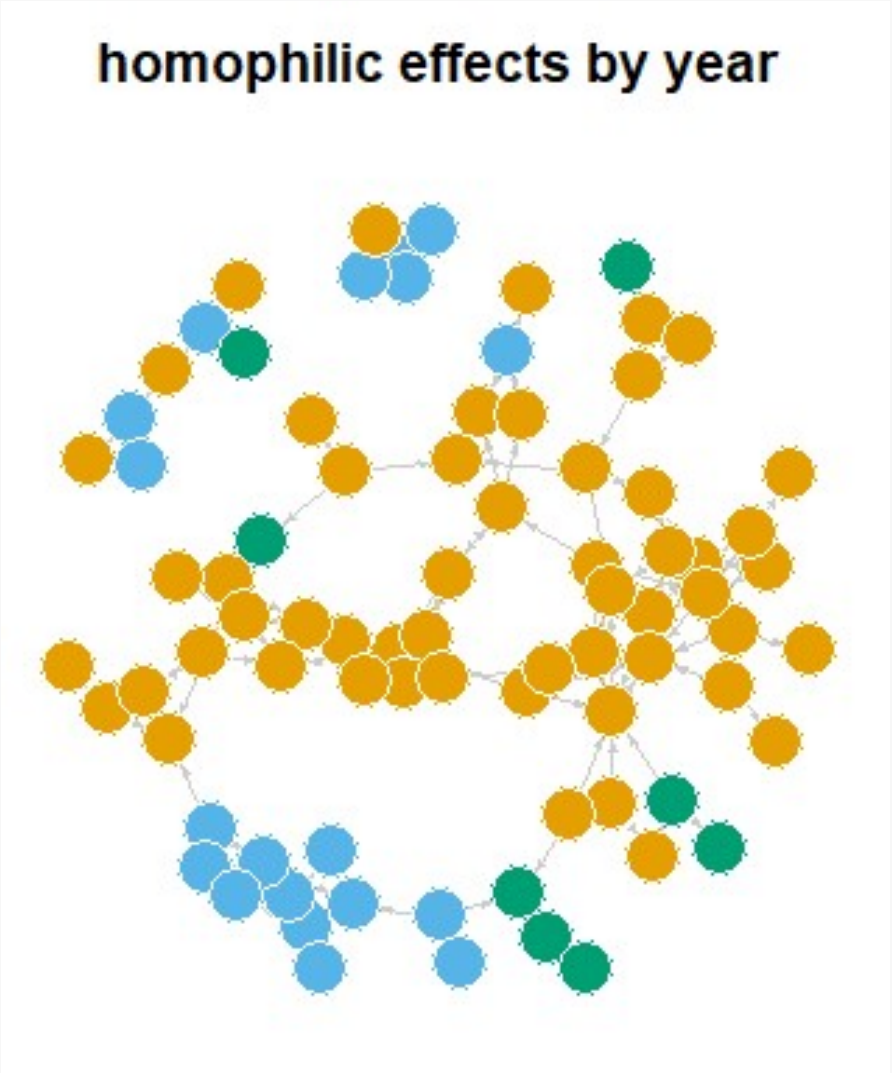
Tutorials plot:
What's the message? Is it clear?



Can you see social dynamics? Can you plot it better to show homophily?


LESS
IS
MORE.
[MORE OR LESS]

Better?



2. Homeplay

What is this network about?

Load the network “UKfaculty”

81 nodes

817 edges

1 nodes attribute (school affiliation)

1 edge attribute (how often they are in contact)

Very simple story

There are people working in a university and they are organized in three groups.

[same as JADS: we have three units]

Story/Goal

You want to be able to show the pattern of contacts within groups and accross groups

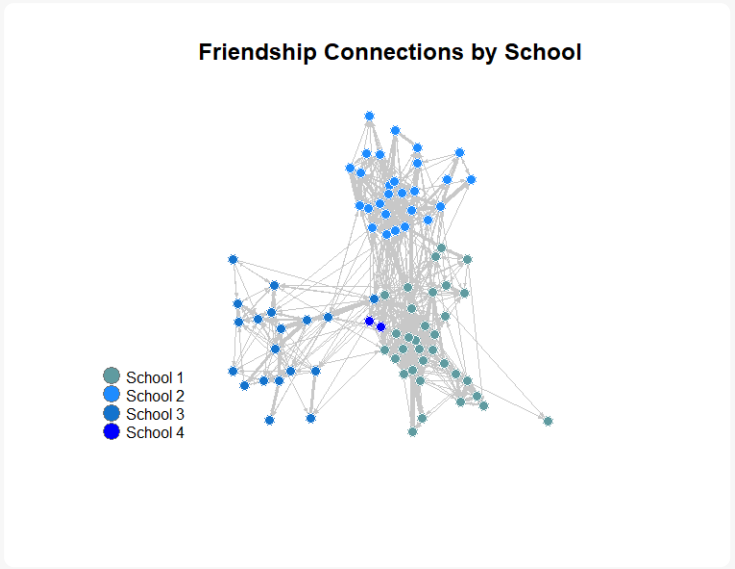
Plot 1 Instruction

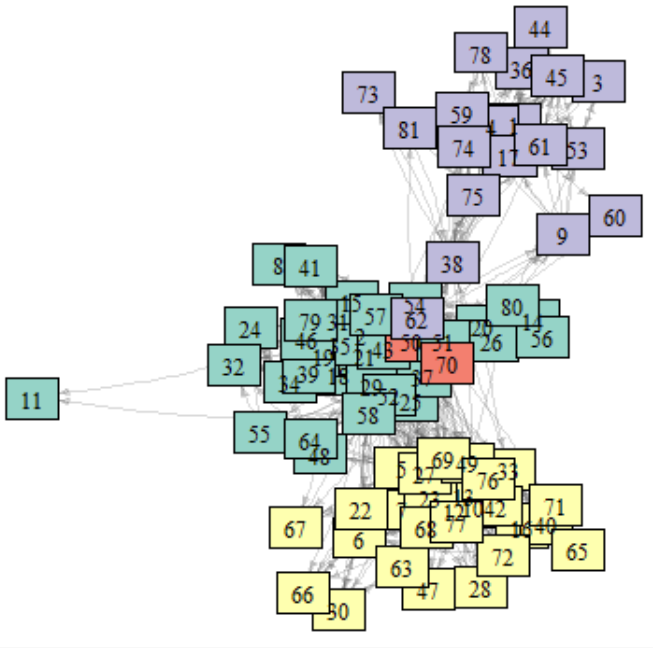
make a visualization using

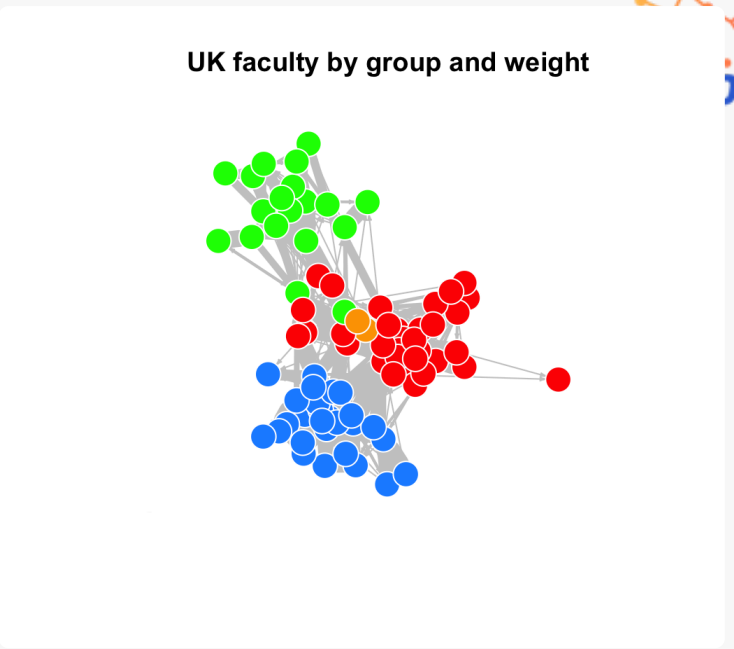
- one edge attribute
- one node attribute

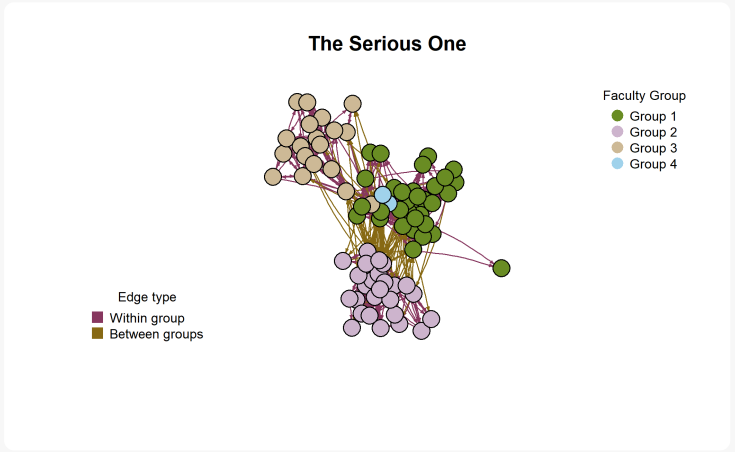
that are able to tell what is happening in this network.

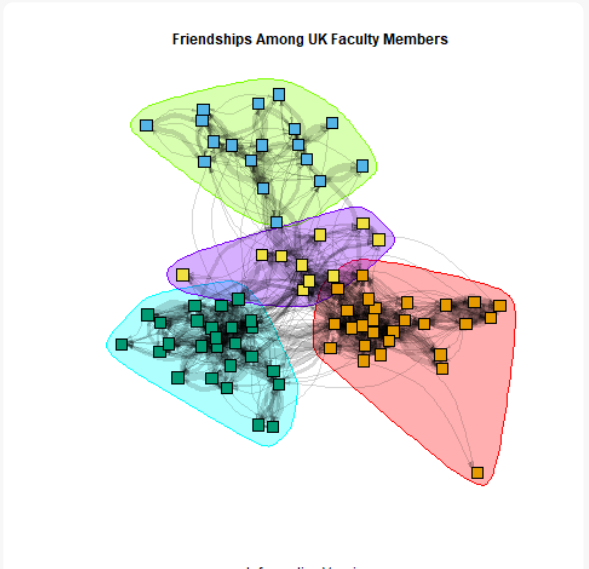
Decide what is the best way to represent this story first and plot after.

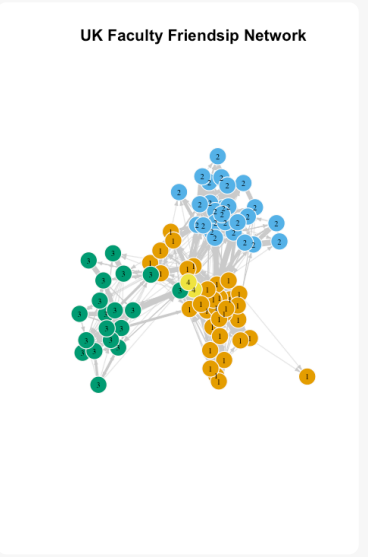

Friendship Connections by School
School 1
School 2
School 3
School 4



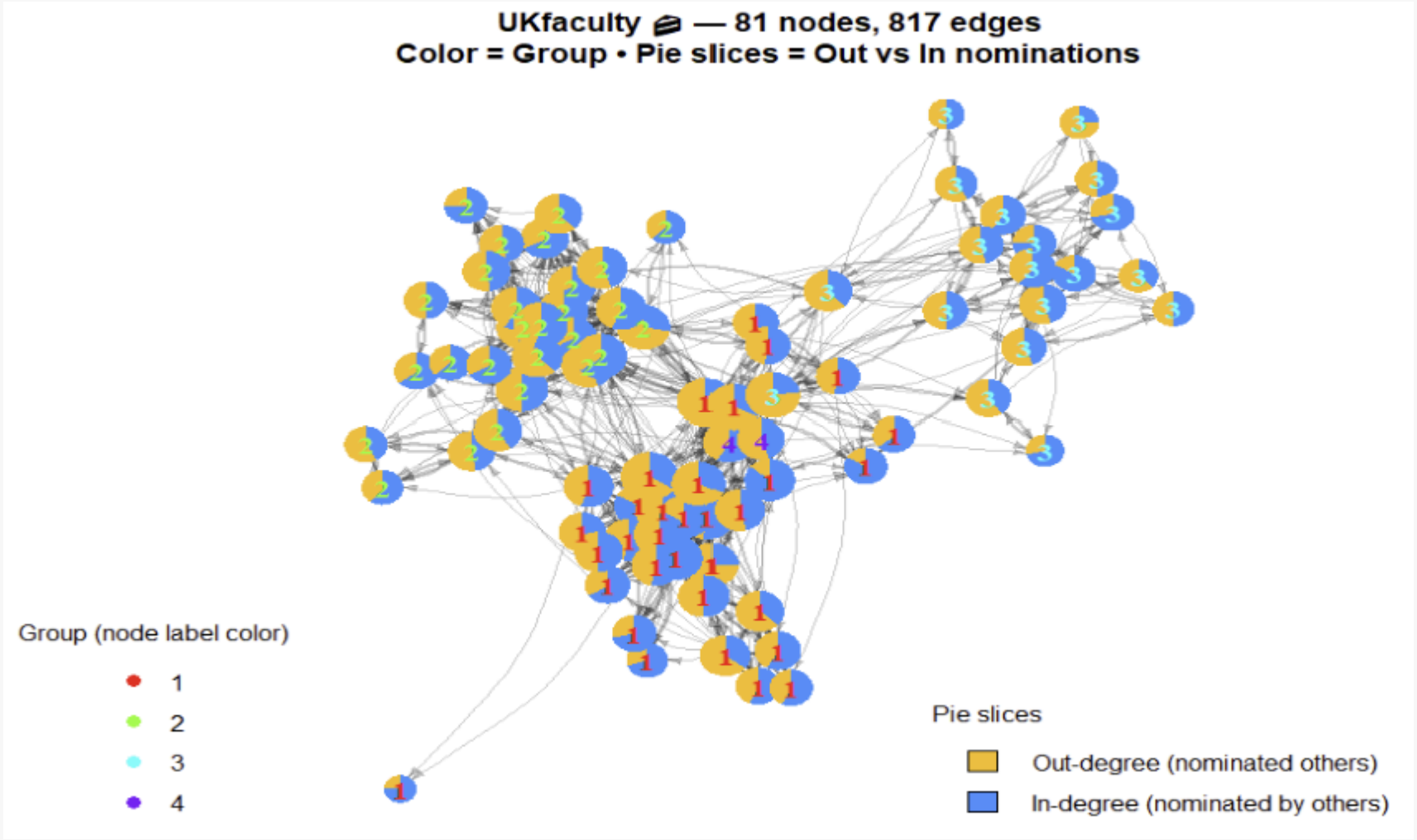

UK faculty by group and weight


The Serious One
Faculty Group
Group 1
Group 2
Group 3
Group 4
Edge type
Within group
Between groups


Friendships Among UK Faculty Members


UK Faculty Friendsip Network

Too powerful for it's own good?

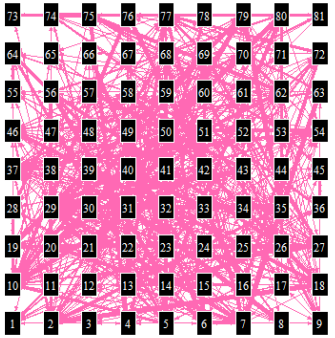


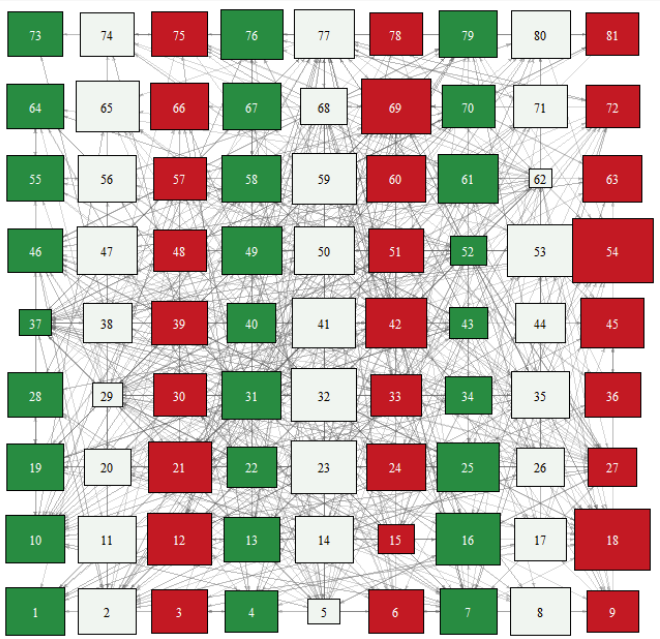
Plot 2 Instruction

Plot the same graph, but with

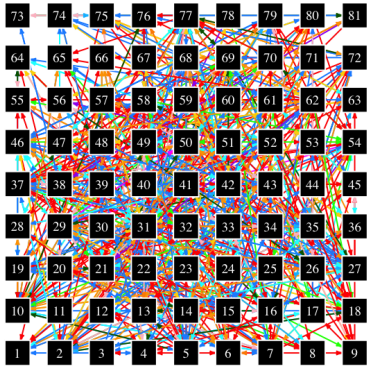
- a lattice layout (just for fun)
- nodes with a rectangular shape
- black nodes
- white nodes labels
- a surprise element of your choice!

Schools: The Best Way to Avoid Meeting Interesting People?

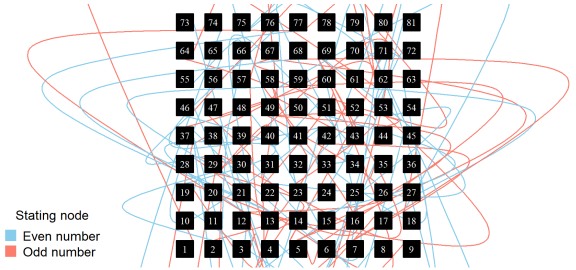




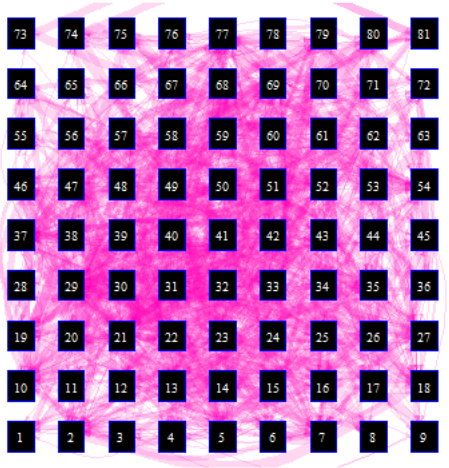
Rainbow edge plot (16 colours)



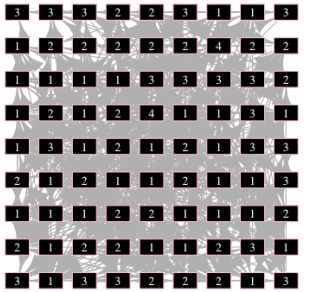
The "Funny" One



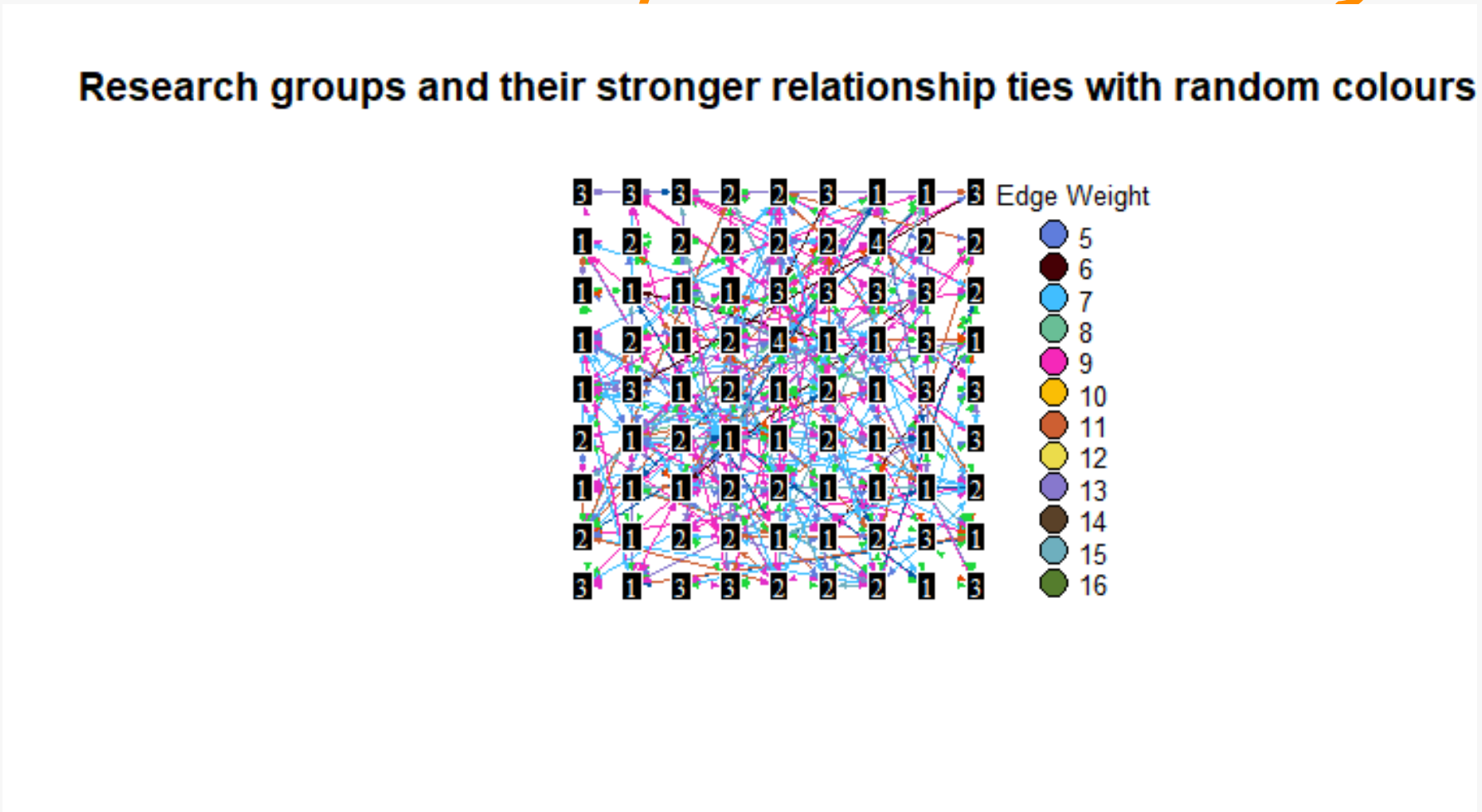
Friendships Among UK Faculty Members



Silly UK Faculty Friendship Network



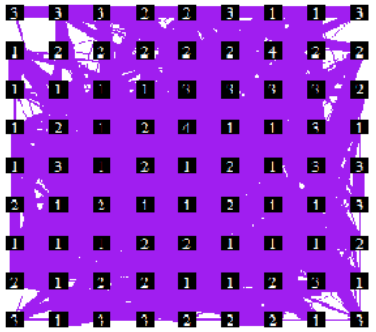
Plot 2 from one of your former colleagues



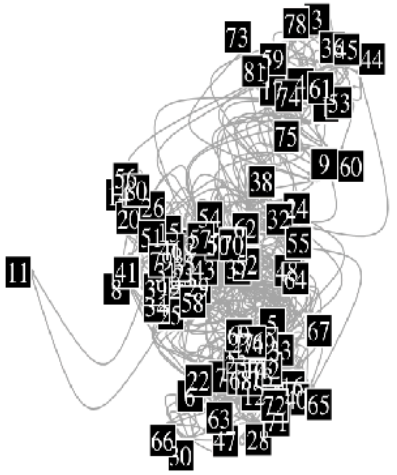
Plot 2 from previous cohorts ^_^

SNA4DS

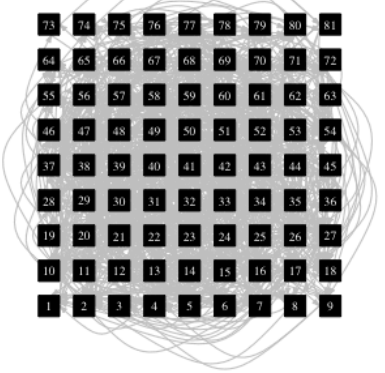
My plot



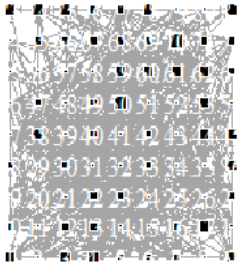
This masterpiece deserves an instant 10 for SNA



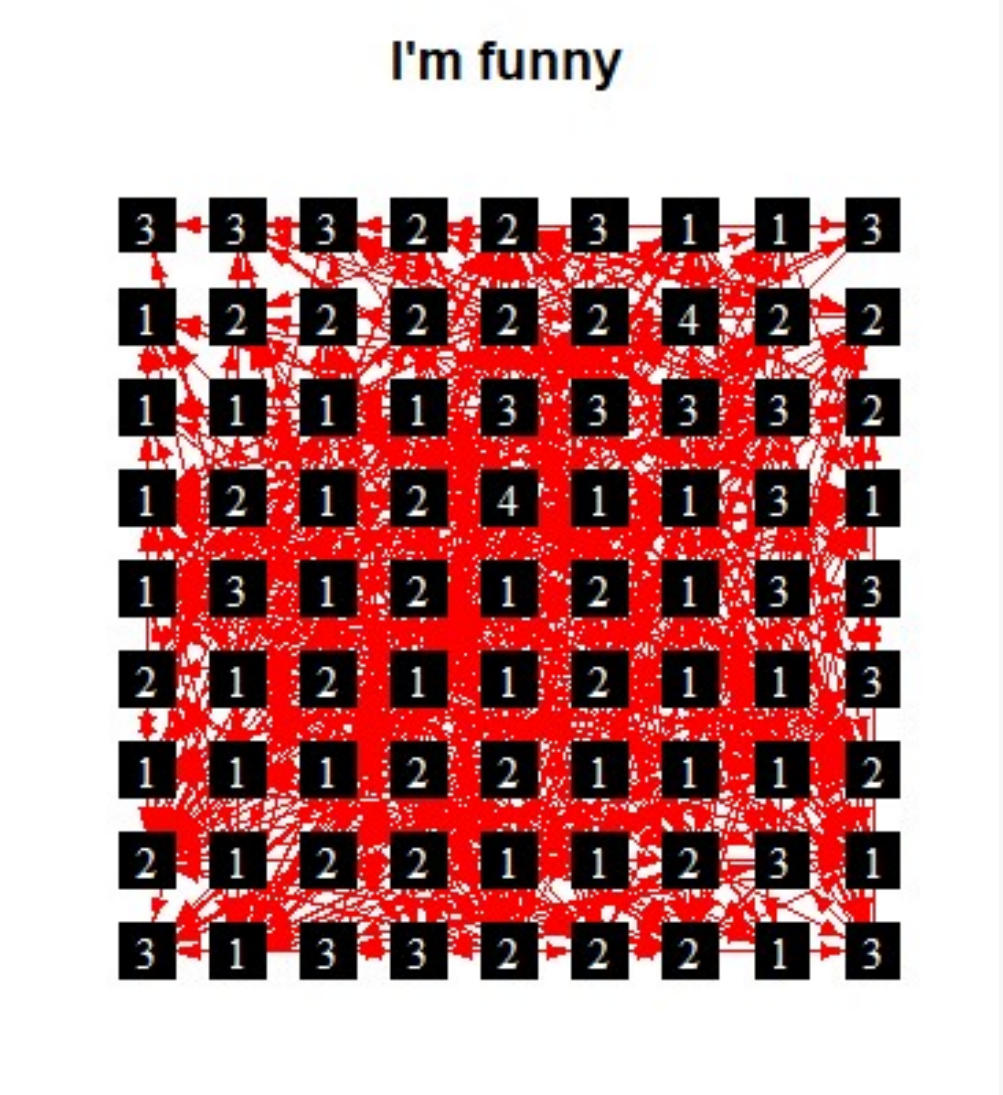
What the h#ll is this network showing



Useless square on an autumn day



I'm funny and I know it!



3. Tricks or treats

Storing the color in the graph

SNA4DS

In the tutorial: node color scale plot

```
# # Generate colors based on how often they hang out:
# colrs <- c("cadetblue", "dodgerblue", "dodgerblue3", "blue", "darkblue" )
# # create a new attribute with the color
# igraph::V(net)$color <- colrs[igraph::V(net)$Hang.Out] # we attribute each category to a colour
#
# plot(net,
#      edge.arrow.size = .2,
#      edge.color = "gray80",
#      vertex.frame.color = "#ffffff",#node perimeter
#      vertex.label = igraph::V(net)$gender,
#      vertex.label.cex = 0.6,  # vertex label size
#      vertex.label.color = "black",
#      vertex.size = igraph::V(net)$DS*2,
# )
```

Storing plot features BEFORE the function

In general, you can also set your preferences before the plot

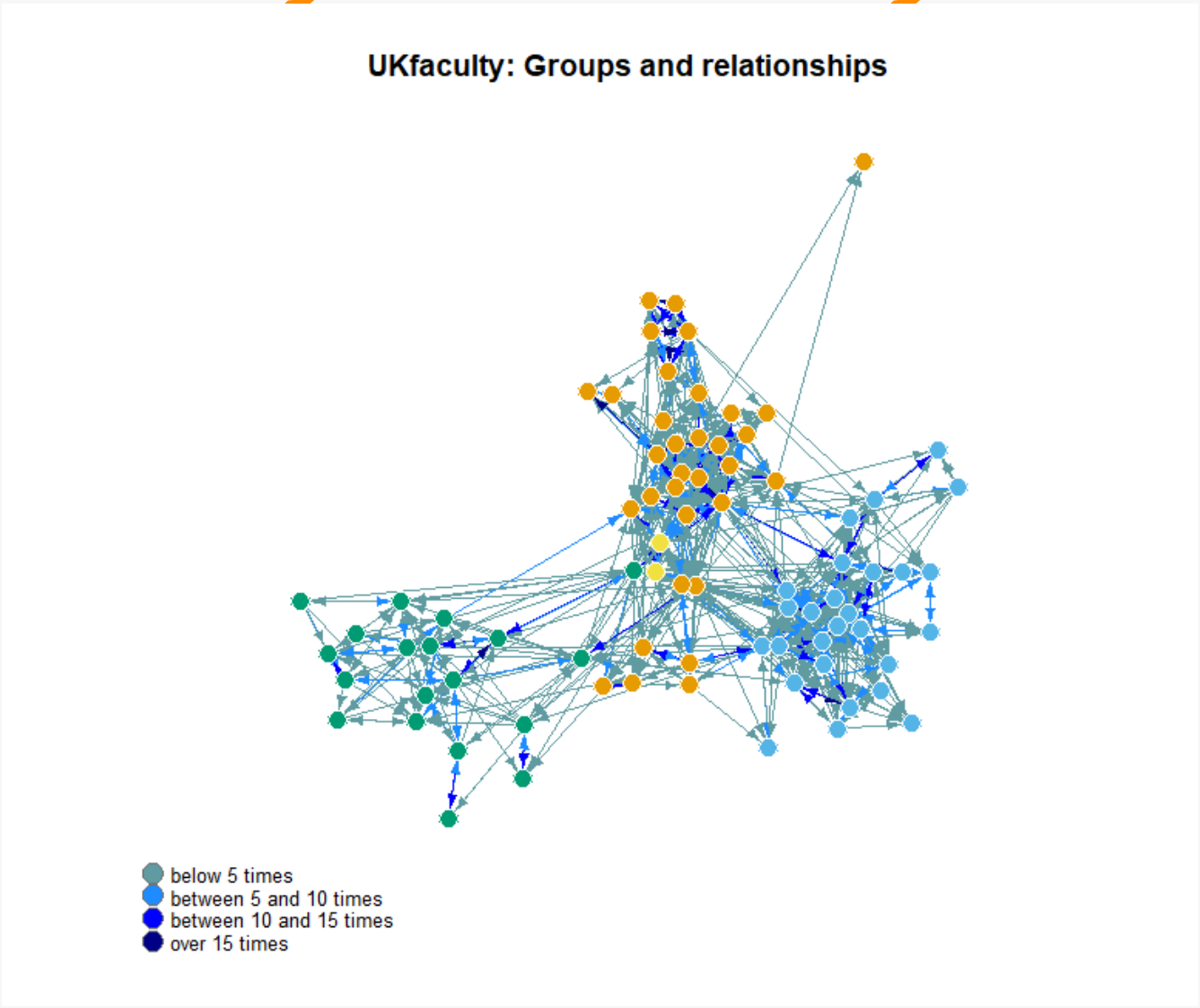
Add attributes with the parameter names the same way you would add columns to a data frame

In the second way parameters remain stored

```
#####################################################################
#                          As we did it so far
#####################################################################
#
# plot(UKfaculty,
#      vertex.shape="circle",
#      vertex.label = NA,
#      vertex.color = igraph::V(UKfaculty)$Group,
#      edge.width = igraph::E(UKfaculty)$weight*0.5,
#      edge.arrow.size = 0.1,
#      layout = igraph::layout_on_grid,
#      main= "Homophilic behavior by group")

#####################################################################
#                           Alternative way
#####################################################################
#
# igraph::V(UKfaculty)$color <-  igraph::V(UKfaculty)$Group
# igraph::E(UKfaculty)$width <-  igraph::E(UKfaculty)$weight*0.5
# igraph::graph_attr(UKfaculty, "layout") <- igraph::layout_on_grid
```

Storing the weight of the edges as color



DIY

SNA4DS

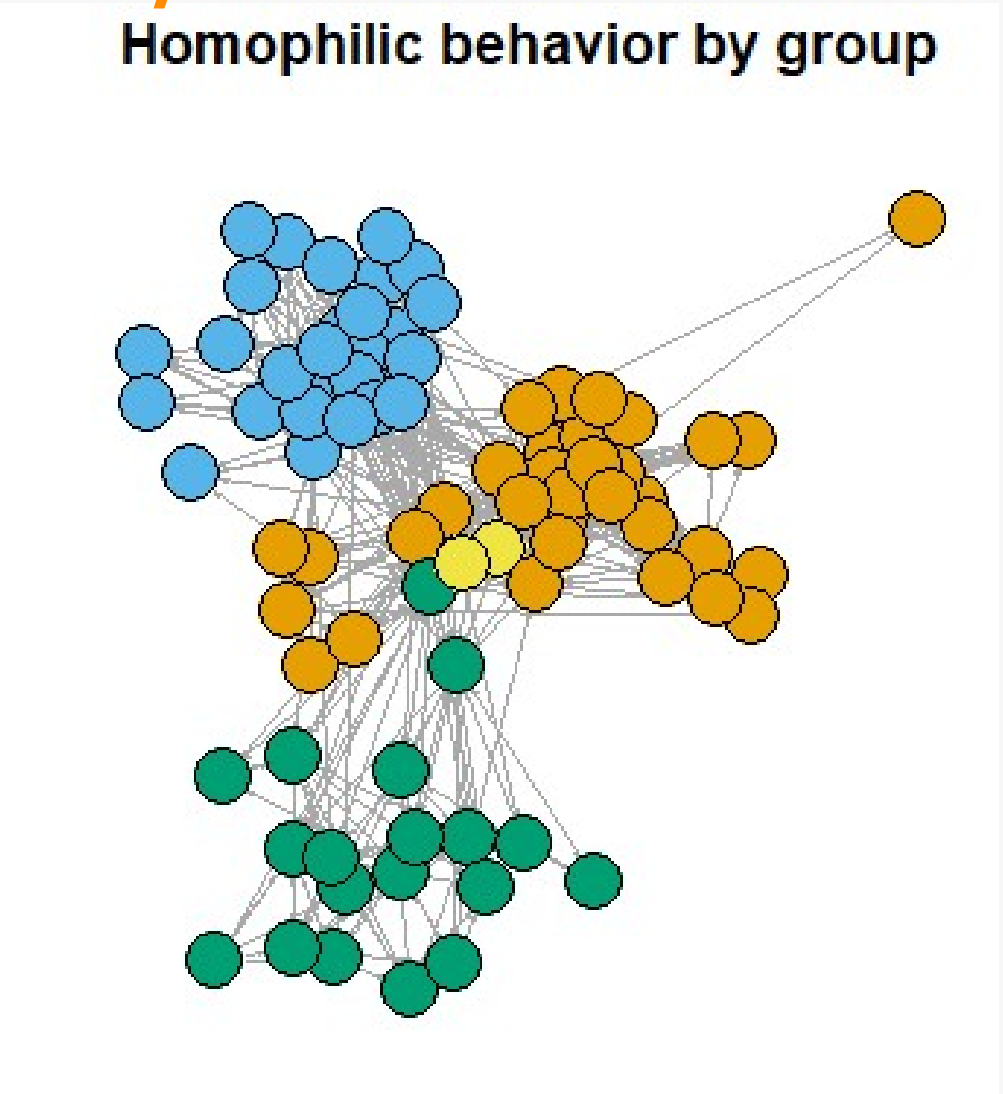
```
###### edge weight in color

# table(igraph::E(UKfaculty)$weight)
#
#
# igraph::E(UKfaculty)$Ecolor <- ifelse(igraph::E(UKfaculty)$weight <= 5, "cadetblue",
#                                       ifelse(igraph::E(UKfaculty)$weight > 5 & igraph::E(UKfaculty)$weig
#                                              ifelse(igraph::E(UKfaculty)$weight > 10 & igraph::E(UKfacul
#                                                     ifelse(igraph::E(UKfaculty)$weight > 15, "darkblue",
#
#
# table(igraph::E(UKfaculty)$Ecolor)
#
#
# plot(UKfaculty,
#      edge.arrow.size = .5,
#      edge.width = 2,
#      edge.color = igraph::E(UKfaculty)$Ecolor,
#      vertex.color = igraph::V(UKfaculty)$Group,
#      vertex.frame.color = "#ffffff",#node perimeter
```

Beyond the standard layout

This can be improved!

```
igraph::layout_with_fr
```



How does the layout work?

The layout in igraph is defined in a matrix with 2 columns and a row for each node.

- The first column indicates its x position
- the second its y position

```
l <-igraph::layout_with_fr(UKfaculty)

head(l)
```

```
##              [,1]     [,2]
## [1,]  2.60265989 3.419298
## [2,] -0.01112223 1.117632
## [3,]  3.61781059 2.482080
## [4,]  2.58279620 2.816784
## [5,] -0.41306693 2.862315
## [6,] -1.86100658 1.872448
```

Modify an existing layout

The scale is not relevant (it is always re-scaled to fit a -1 to 1 plotting area.

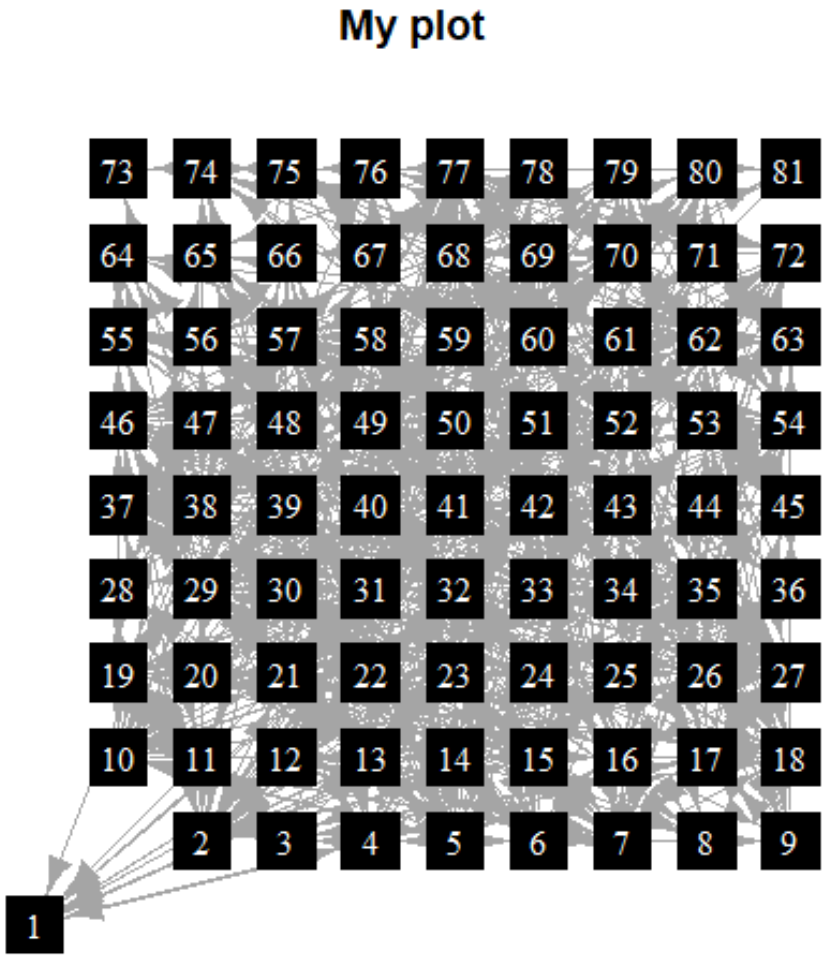
You can get this layout before plotting by just calling the layout function on the graph:

```
l <- igraph::layout_on_grid(UKfaculty)

l[1, ] <- c(-1, -1)

# plot(UKfaculty,
#       vertex.shape="rectangle",
#       vertex.label.color = "white",
#       vertex.color = "black",
#       layout = l,
#       main= "My plot")
```

Beyond the layout



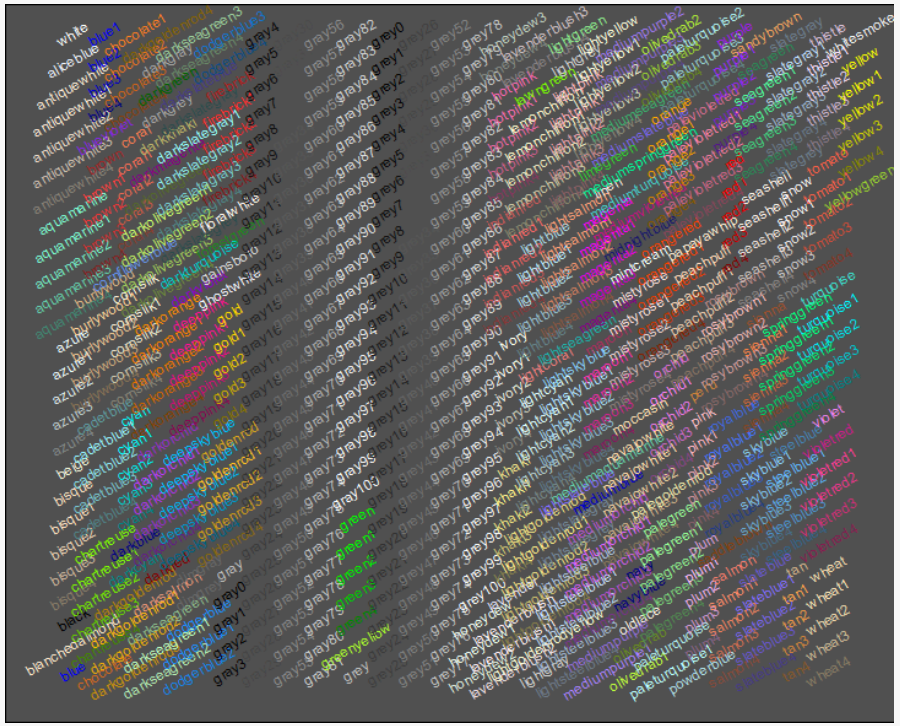
Move!

This cannot work in the slides!

```
# igraph::tkplot(UKfaculty)
```

Color palette

- hexadecimal color codes
- `demo("colors")` that are inbuilt in R `grDevices`
- several packages



4. Final Remarks

Your Book

... is plotting using `network`

that code won't work with `igraph` objects.

Still, the logic is very similar

`snafun::to_igraph` & `snafun::plot`

OR you can always use `intergraph` and plot in `igraph`

OR look up the corresponding paramenters and plot in `network`

Parameters RECAP there are many more. These are the essentials



vertex:

- `vertex.shape`
- `vertex.color`
- `vertex.label`
- `vertex.size`
- `vertex.frame.color`

edge:

- `edge.arrow.size`
- `edge.color`
- `edge.width`

overall: `layout`

Some more JADS students Viz!

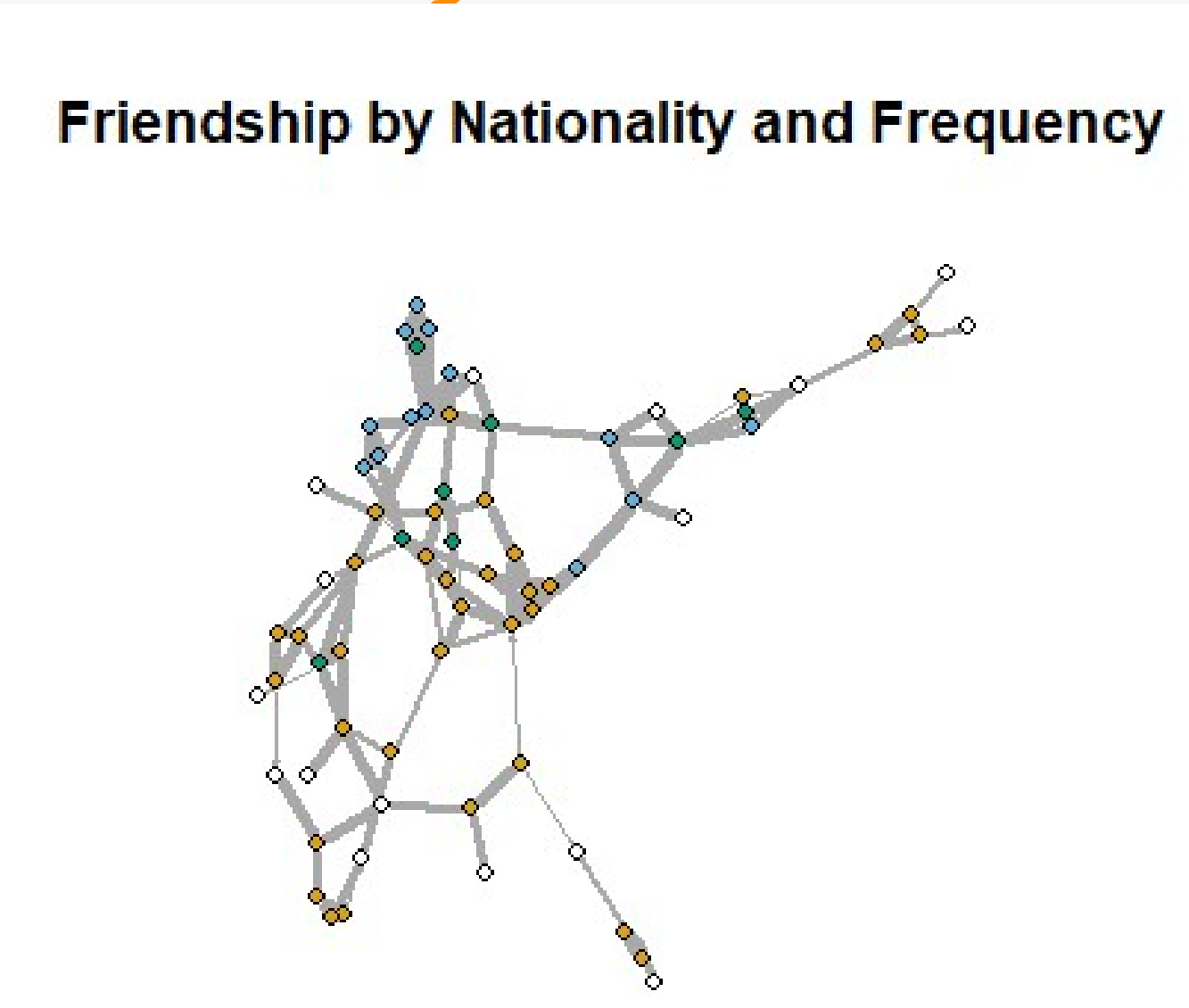


Tutorial data from an earlier cohort

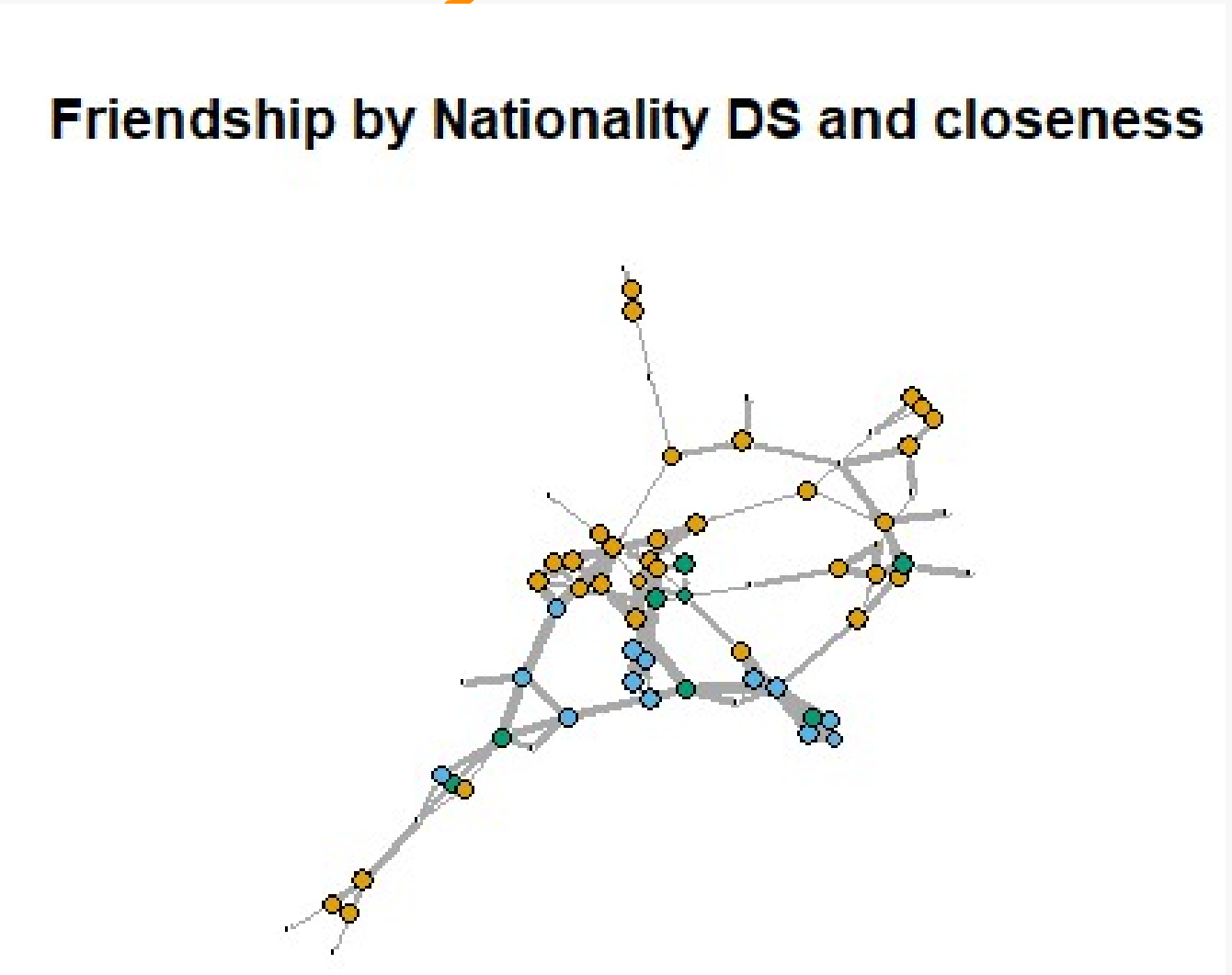
Still...

Let's take a look to JADS data!

What does this image tell us?



What does this image tell us?



Etenstijd!

